

ADVANCED MACHINE LEARNING · WORKSHEET 12

Regularization (L1 and L2)

Overfitting · Ridge Regression · Lasso Regression

STUDENT COPY — Lecture 12

Name:

Roll No.:

Date:

PART I: THE PROBLEM OF OVERFITTING

Introduction: Why do we need regularization?

▷ HOOK

When a model fits every noisy point, the slope becomes abnormally large, and it fails to generalize. Regularization prevents this by constraining coefficients.

The overfitting problem

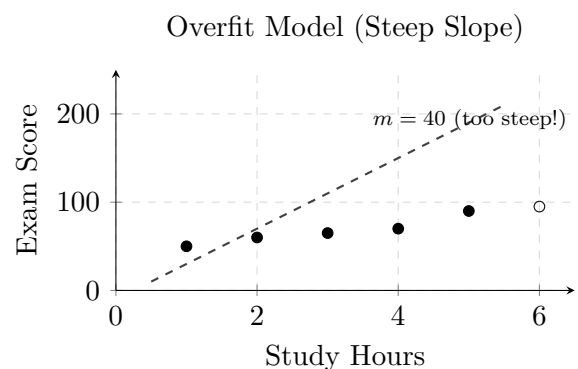
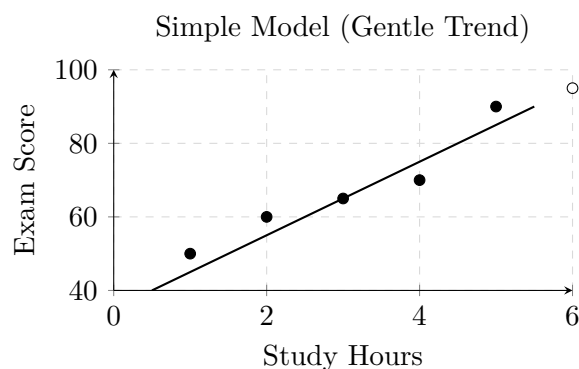
When building machine learning models (Linear Regression, Logistic Regression, etc.), we often see a model perform exceptionally well on training data but fail to generalize to unseen test data. This is called _____.

An overfitted model has: High _____, Low _____. It memorizes the noise and exact patterns of training data rather than capturing the underlying distribution.

(Hint: recall the Bias-Variance Tradeoff from Lecture 9.)

Why fitting every point is dangerous

Quick Example: A simple model draws a gentle trend. An overfit model contorts to hit every point, leading to absurd predictions on new data.



This is why we need a mechanism that says: "Don't try too hard to fit every point."

$$\hat{y} = w_0 + w_1 x_1 + w_2 x_2 + \dots + w_p x_p$$

$w_0 = \text{Intercept}$ $w_i \rightarrow \text{coefficients / weights}$
 $(1 \leq i \leq p)$

$$\boxed{\frac{\partial \hat{y}}{\partial x_j} = w_j}$$

Case 1: $w_j = 0.05$ and $\Delta x_j = 0.01 \Rightarrow \Delta y = 0.0005$
 $\hookrightarrow$ Very less sensitive to the changes in the data
 (Underfitting)

Case 2: $w_j = 1000$ and $\Delta x_j = 0.01 \Rightarrow \Delta y = 10$

$\hookrightarrow$ Highly sensitive to any change

$\hookrightarrow$ Overfitting (Model is highly flexible)

Case 3: $w_j = 0.5$ and $\Delta x_j = 0.01 \Rightarrow \Delta y = 0.05$

$\hookrightarrow$ Optimal complexity

Regularization $\Rightarrow$ Reduces overfitting by improving generalization.

Even though training performance can decrease but validation performance / testing performance will increase.

Types of Regularization

- 1. $L1$ & $L2$, Elastic Net } Penalty Based Methods
- 2. Early Stopping
- 3. Dropout Regularization

⋮

$L2$ Regularization (Ridge Regression)

for Simple L.R case: $\hat{y} = mx + c$

$$L_{\text{Ridge}} = \sum_{i=1}^n (y_i - \hat{y}_i)^2 + \lambda m^2$$

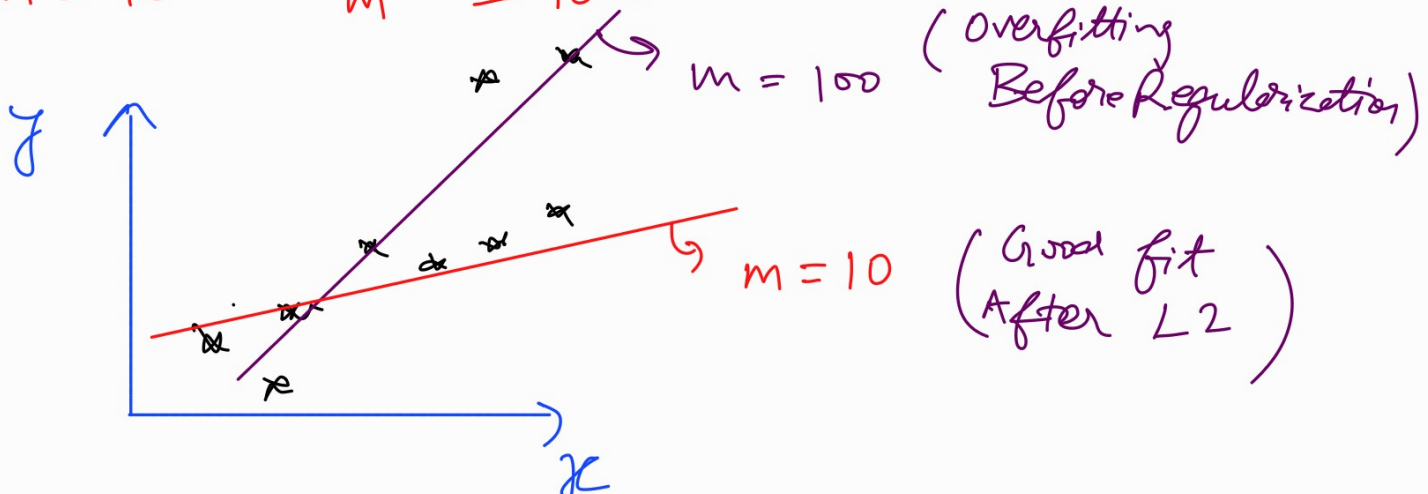
$$L_{\text{Ridge}} = \sum_{i=1}^n (y_i - mx - c)^2 + \lambda m^2$$

$\lambda \rightarrow$ Hyperparameter

$\hookrightarrow$ Strength of Regularization

Penalty

$$m = 100 \Rightarrow m^2 = 10000$$



For $m=100 \Rightarrow$ Training Performance $= 0.85$
Validation $= 0.70$

For $m=10 \Rightarrow$ Training Perf. $= 0.80$
Validation Perf. $= 0.75$ } Bias $\uparrow$
and Var $\downarrow$

If $\lambda = 0 \Rightarrow$ Overfitting (Normal OLS Solution)

If $\lambda \rightarrow \infty \Rightarrow m \rightarrow 0$ (Underfitting)
↳ But not exactly zero

For optimal $\lambda \Rightarrow$ Good fit Model.

Closed form solⁿ for Simple Ridge Regression

$$L_{\text{Ridge}} = \sum_{i=1}^n (y_i - mx_i - c)^2 + \lambda m^2$$

$$\frac{\partial L_{\text{Ridge}}}{\partial c} = 0 \quad \text{and} \quad \frac{\partial L_{\text{Ridge}}}{\partial m} = 0$$

$$\frac{\partial L_{\text{Ridge}}}{\partial c} = -2 \sum_{i=1}^n (y_i - mx_i - c) = 0$$

$$\sum_{i=1}^n (y_i - mx_i - c) = 0 \Rightarrow \boxed{c = \bar{y} - m\bar{x}}$$

$$L_{\text{Ridge}} = \sum_{i=1}^n (y_i - mx_i - \bar{y} + m\bar{x})^2 + \lambda m^2$$

$$L_{\text{Ridge}} = \sum_{i=1}^n \{(y_i - \bar{y}) - m(x_i - \bar{x})\}^2 + \lambda m^2$$

$$\frac{dL_{\text{Ridge}}}{dm} = -2 \left[\sum_{i=1}^n \{(y_i - \bar{y}) - m(x_i - \bar{x})\} \{x_i - \bar{x}\} \right] + 2\lambda m = 0$$

$$- \left[\sum_{i=1}^n \{(y_i - \bar{y})(x_i - \bar{x}) - m(x_i - \bar{x})^2\} \right] + \lambda m = 0$$

$$- \sum_{i=1}^n (y_i - \bar{y})(x_i - \bar{x}) + m \sum_{i=1}^n (x_i - \bar{x})^2 + \lambda m = 0$$

$$m \left[\sum_{i=1}^n (x_i - \bar{x})^2 + \lambda \right] = \sum_{i=1}^n (y_i - \bar{y})(x_i - \bar{x})$$

$$m = \frac{\sum_{i=1}^n (y_i - \bar{y})(x_i - \bar{x})}{\sum_{i=1}^n (x_i - \bar{x})^2 + \lambda}$$

$$\lambda \uparrow \Rightarrow m \downarrow$$

When $\lambda \rightarrow \infty$, $m \rightarrow 0$ but not exactly zero

No. of features will remain same.

Ridge Regression for m-dimensions

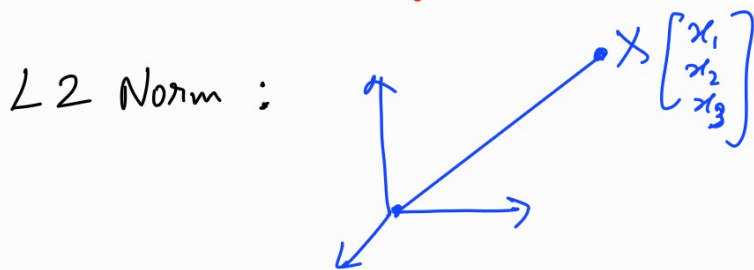
$$L_{\text{ridge}} = \sum_{i=1}^n (y_i - \hat{y}_i)^2 + \lambda \sum_{j=1}^p w_j^2$$

Penalty is of L2 Norm.

$$\hat{y}_i = w_0 + w_1 x_1 + w_2 x_2 + \dots + w_p x_p$$

$$w^* = (X^T X + \lambda I)^{-1} X^T y$$

$I \rightarrow$ Identity matrix



$$\sqrt{x_1^2 + x_2^2 + x_3^2} = \text{L2 Norm}$$

L1 Norm : Manhattan distance $|x_1| + |x_2| + |x_3| \Rightarrow \underline{\text{L1 Norm}}$

fill feedback

Lasso Regression (L1 Regularization)

$$L_{\text{Lasso}} = \sum_{i=1}^n (y_i - \bar{y}_i)^2 + \lambda \sum_{j=1}^p |w_j|$$

↳ Not smoothly differentiable

↳ Penalty of L1 Norm

↳ Closed form solⁿ does not exist.

L1 Regularization for Simple L.R

$$L_{\text{Lasso}} = \sum_{i=1}^n (y_i - mx_i - c)^2 + \lambda |m|$$

Case 1: $m > 0 \Rightarrow$

$$m = \frac{\sum_{i=1}^n (y_i - \bar{y})(x_i - \bar{x}) - \frac{\lambda}{2}}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

When $\frac{\lambda}{2} = \sum_{i=1}^n (y_i - \bar{y})(x_i - \bar{x})$ then $\boxed{m=0}$

Because of this nature it can work as Feature Selection Method (Embedded Methods) because it will make some coefficients exactly equal to zero.
 $\Rightarrow$ No. of features are reducing.

$\Downarrow$
Sparsity (Too many Zeros)

Geometric intuition in Linear Regression

In 2D space, an overfitted model contorts to pass through every point, becoming hypersensitive to tiny variations in x . The slope m approaches infinity.

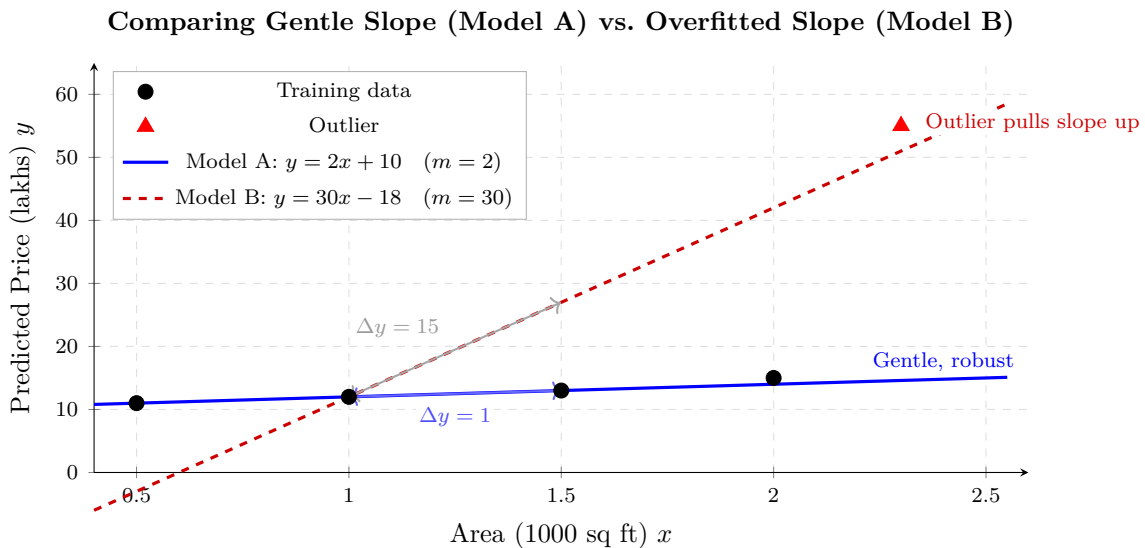
Conclusion: Abnormally high coefficient values (steep slopes) are strong indicators of overfitting. A slight change in the test data x will result in a massive change in the prediction $\hat{y}$, leading to high variance and significant test errors.

Seeing coefficient explosion in action

Compare two models predicting house price (in lakhs):

	Model A: $y = 2x + 10$	Model B: $y = 30x - 18$
Prediction at $x = 1.0$	$2(1.0) + 10 = 12$	$30(1.0) - 18 = 12$
Prediction at $x = 1.5$	$2(1.5) + 10 = 13$	$30(1.5) - 18 = 27$
Change from $\Delta x = 0.5$	$\Delta y = 1$	$\Delta y = 15$

Model B is highly sensitive, demonstrating high variance due to a large slope.



Question: What does a slope of $m = 4500$ indicate about generalization?

★ KEY INSIGHT

Large coefficient values are the symptom of overfitting. The model is hypersensitive to small changes in input, producing wildly different predictions. We need a mechanism to control these coefficient magnitudes.

What is regularization?

Regularization is a technique used to prevent overfitting by adding a penalty term to the loss function. This penalty discourages the model from using unnecessarily large coefficients.

Regularization is the fundamental approach for controlling parametric models like Linear Regression, Logistic Regression, and Perceptrons.

Types of regularization

Method	Core Mechanism / Description
Ridge Regression (L2 Regularization)	Constrains coefficient magnitudes (prevents exploding slopes)
Lasso Regression (L1 Regularization)	Constrains coefficients and performs automatic feature selection
Elastic Net	Combines both L1 and L2 regularization mechanisms

(Elastic Net is out of scope of this lecture.)

◆ TAKEAWAY

Overfitting shows up as abnormally large coefficients. Regularization prevents this by adding a penalty term to the loss function, forcing the model to balance fitting the data against keeping coefficient magnitudes small.

PART II: L2 REGULARIZATION — RIDGE REGRESSION

Ridge Regression: Geometric and Mathematical Intuition (2D)

▷ HOOK

To control the abnormally high slope m characteristic of an overfitted model, Ridge Regression strategically modifies the standard loss function. Instead of only minimizing the error, the optimizer must also pay a price proportional to the square of the slope.

Modifying the cost function

For Ordinary Least Squares (OLS) linear regression, the cost function is the Sum of Squared Errors (SSE):

$$L_{\text{OLS}} = \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

We know that abnormally large coefficients cause severe overfitting and high variance. If we want the optimization algorithm to avoid picking large coefficients, what modification should we make to the loss function?

- (A) Subtract the coefficient values from the loss function
- (B) Add a penalty proportional to the squared coefficients to the loss function
- (C) Multiply the loss function by the total number of features

Your choice: _____ Why? _____

This is exactly what Ridge Regression does.

In Ridge Regression, we append a regularization penalty term:

$$L_{\text{Ridge}} = \sum_{i=1}^n (y_i - mx_i - c)^2 + \lambda m^2$$

Here, $\lambda \geq 0$ is a hyperparameter that controls the intensity of the penalty applied to the slope m .

The mechanism of the penalty (Bias–Variance Tradeoff)

Why does adding λm^2 improve generalization? Let's analyze step-by-step:

1. In an unregularized model ($\lambda = 0$), minimizing training error drives OLS error to 0, but pushes slope m to very large values.
2. In Ridge Regression ($\lambda > 0$), if m becomes large, the penalty term λm^2 explodes, making the overall loss L_{Ridge} high.

Closed-Form Derivation for 2D Ridge Regression

To see exactly how λ controls m , let's derive the slope formula from scratch by minimizing L_{Ridge} :

Step 1: Partial derivative with respect to intercept (c)

Setting $\frac{\partial L_{\text{Ridge}}}{\partial c} = 0$:

$$\sum_{i=1}^n -2(y_i - mx_i - c) = 0 \implies c = \bar{y} - m\bar{x}$$

Step 2: Substitute $c = \bar{y} - m\bar{x}$ back into loss function

Substituting c back into L_{Ridge} eliminates the intercept, expressing the loss purely in terms of slope m :

$$L_{\text{Ridge}}(m) = \sum_{i=1}^n [(y_i - \bar{y}) - m(x_i - \bar{x})]^2 + \lambda m^2$$

Step 3: Differentiate with respect to slope m

Taking the derivative $\frac{dL_{\text{Ridge}}}{dm} = 0$:

$$-2 \sum_{i=1}^n (x_i - \bar{x}) [(y_i - \bar{y}) - m(x_i - \bar{x})] + 2\lambda m = 0$$

Divide by 2 and expand:

$$-\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) + m \sum_{i=1}^n (x_i - \bar{x})^2 + \lambda m = 0$$

Grouping terms with m :

$$m \left(\sum_{i=1}^n (x_i - \bar{x})^2 + \lambda \right) = \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$$

$$m_{\text{Ridge}} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2 + \lambda}$$

Connecting the Derivation to Bias and Variance

Now that we have derived the exact formula $m_{\text{Ridge}} = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2 + \lambda}$, we can understand points 3 and 4 with mathematical clarity:

3. **Compromise:** Notice that λ sits directly in the **denominator**. As λ increases, the denominator grows larger while the numerator remains fixed. To minimize L_{Ridge} , the optimizer willingly increases the OLS error (sacrificing a small amount of training accuracy, i.e., introducing slight **Bias**) in order to drastically shrink m .
4. **Result:** A smaller slope m yields a flatter, smoother line. The model becomes far less sensitive to random fluctuations or outliers in training data, drastically reducing **Variance**. This optimizes the Bias–Variance Tradeoff, leading to superior generalization on unseen test data.

Based on the mechanism above, fill in how the compromise works:

What happens	Effect on Bias	Effect on Variance	Net result
m is shrunk toward zero	☐ Increases ☐ Decreases	☐ Increases ☐ Decreases	—

(Hint: Think about what a flatter line means — does it follow training points closely, or does it generalize better?)

★ KEY INSIGHT

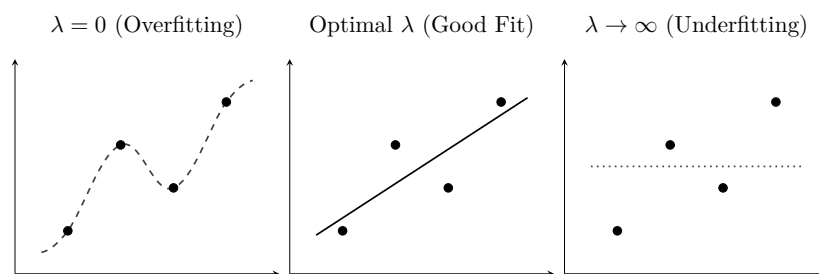
Ridge Regression forces a compromise: it willingly increases bias slightly in order to dramatically reduce variance. The net effect is better generalization to unseen data.

Visualizing the effect of λ

Using a real-world dataset, tuning λ demonstrates three regimes:

λ value	Slope behavior	Regime
$\lambda = 0$	High slope; model = standard OLS	_____
Optimal λ	Moderate slope; main trend captured	Good fit
$\lambda \rightarrow \infty$	$m \approx 0$; essentially flat line	_____

(Hint: Which regime from Lecture 9 corresponds to high slope? Which to a flat line?)



◆ TAKEAWAY

Ridge Regression modifies the OLS loss by adding λm^2 . The optimization must compromise between fitting the training data and keeping coefficients small: slight increase in bias, large decrease in variance.

Multiple Ridge Regression (N-Dimensional)

For the general case with p features, the Ridge Normal Equation is:

$$\beta_{\text{ridge}} = (X^T X + \lambda I)^{-1} X^T y$$

(The derivation of this equation is out of scope of this lecture.)

Note: The single-feature (2D) example above is used for mathematical simplicity, while the practical benefits of Ridge Regression become most apparent in multi-feature regression ($N \geq 2$).

The unregularized OLS solution is: $\beta_{\text{OLS}} = (X^T X)^{-1} X^T y$.

Observation: The term λI added to $X^T X$ ensures the matrix is always invertible. This is why Ridge handles **multicollinearity** well — even when $X^T X$ is singular or near-singular, $(X^T X + \lambda I)$ is invertible for any $\lambda > 0$.

◆ TAKEAWAY

In 2D, Ridge adds λ to the denominator of the slope formula, shrinking m toward zero. In N dimensions, the Normal Equation becomes $(X^T X + \lambda I)^{-1} X^T y$, which always exists because λI

guarantees invertibility.

Key theoretical properties of Ridge Regression

▷ HOOK

Understanding how Ridge behaves as λ changes is essential for interviews and real applications. These properties form the core of advanced ML understanding.

4.1 The Shrinkage Effect: Do coefficients ever reach absolute zero?

As $\lambda \rightarrow \infty$, what happens to the feature coefficients $(w_1, w_2, \dots)$?

All coefficients shrink asymptotically towards zero. However, in Ridge Regression (L2 penalty), the coefficients will **NEVER** become exactly zero. They become infinitesimally small (e.g., 10^{-5}) but remain non-zero.

Ridge will always use all input features, no matter how heavily regularized it is.

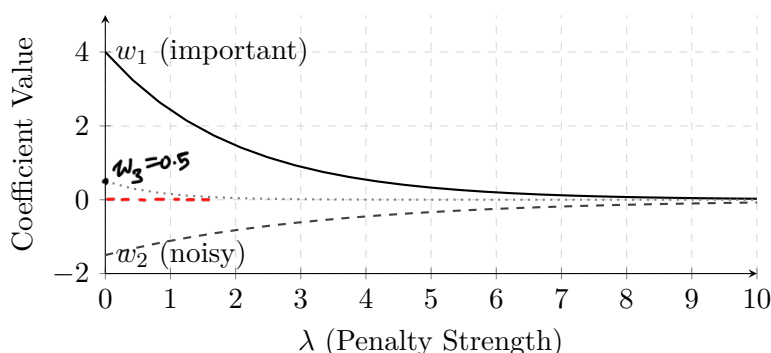
4.2 Asymmetric Shrinkage: Which coefficients shrink faster?

Suppose three features have unregularized OLS coefficients $w_1 = 5000$, $w_2 = 1000$, and $w_3 = 1$. Because the penalty term is λw^2 :

Coefficient	Contribution to penalty
$w_1 = 5000$	$25,000,000 \cdot \lambda$
$w_2 = 1000$	$1,000,000 \cdot \lambda$
$w_3 = 1$	$1 \cdot \lambda$

The larger the initial coefficient, the faster and more aggressively it is penalized. The optimizer immediately targets w_1 for massive reduction. For very high λ , all coefficients converge closely together near zero.

What does this look like visually? If you plot each coefficient's magnitude (y-axis) against λ (x-axis) — called a *regularization path* — the largest coefficients drop steeply first while smaller ones barely move. As λ grows, all curves converge toward zero but never touch the x-axis.

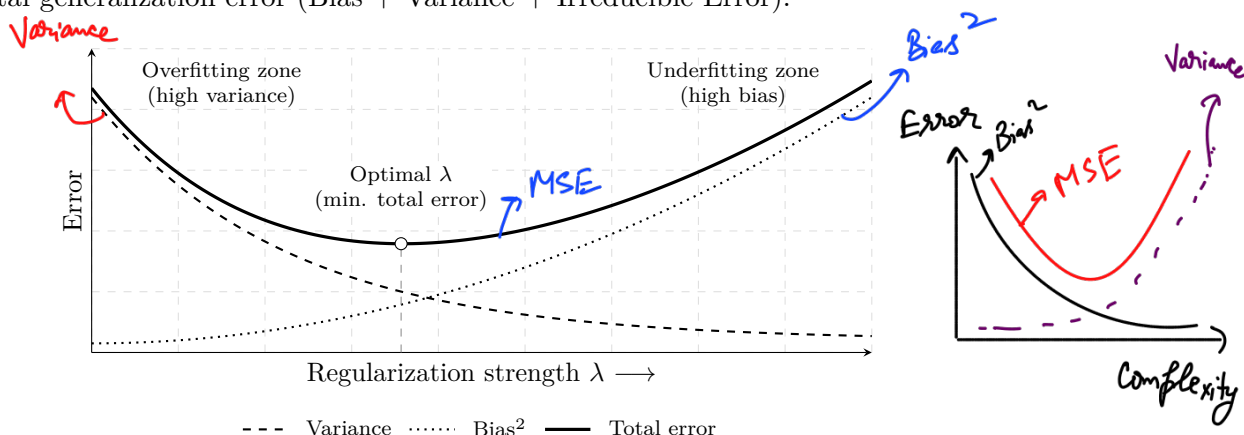


4.3 The Impact on Bias and Variance

Regularization is the direct manipulation of the Bias–Variance Tradeoff.

- **Low λ (e.g., $\lambda \rightarrow 0$):** Coefficients are unrestricted. Bias: Low. Variance: High. The model overfits and fluctuates wildly on test data.
- **High λ (e.g., $\lambda \rightarrow \infty$):** Coefficients are restricted severely, drawing an almost flat line. Bias: High. Variance: Low. The model underfits, basically outputting the mean.

The Sweet Spot: The goal of hyperparameter tuning (via Cross-Validation) is to find the intermediate λ where Variance has dropped significantly but Bias has only increased slightly — minimizing total generalization error ($\text{Bias}^2 + \text{Variance} + \text{Irreducible Error}$).



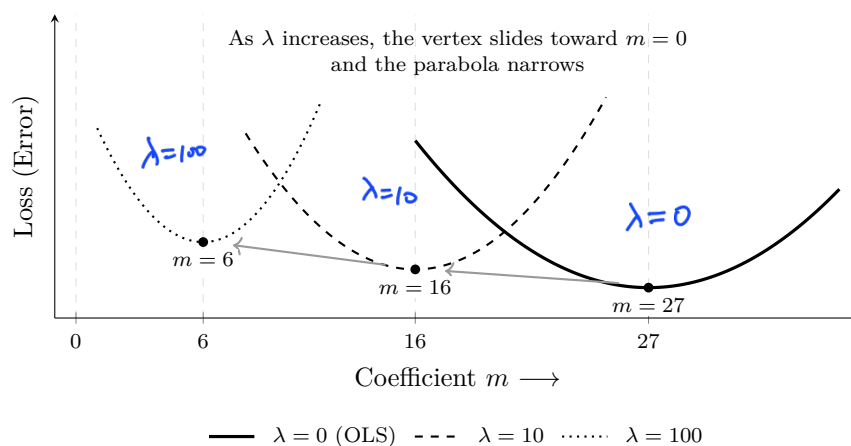
4.4 The Geometric Shifting of the Loss Function

For OLS ($\lambda = 0$), the 2D loss function is a parabola minimized at a large coefficient magnitude.

As λ increases:

1. **The parabola shifts:** The vertex (minimum) moves horizontally from $m = 27$ toward $m = 0$.
2. **The parabola narrows:** The curve becomes steeper or "tighter."
3. **The new minimum:** Because the landscape is translated toward the origin, Gradient Descent converges at a smaller coefficient value.

For a 3D contour plot with two weights (w_1, w_2), the center of the elliptical contours shifts diagonally toward the origin $(0, 0)$ as λ increases.



4.5 Why is it called "Ridge" Regression? (Hard Constraint Intuition)

Note: This section is optional self-reading for interested students. It will not be covered in the lecture and is not part of the examination.

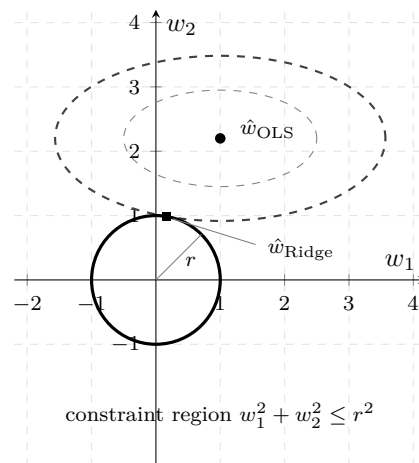
Until now, we have viewed Ridge as a **Soft Constraint** — adding a penalty directly to the loss. Mathematically, this is equivalent to solving OLS subject to a **Hard Constraint**:

$$\text{Minimize } (XW - Y)^T (XW - Y) \quad \text{subject to} \quad w_1^2 + w_2^2 + \dots + w_n^2 \leq r^2$$

The geometric picture:

- In a 2D parameter space (w_1, w_2) , the constraint $w_1^2 + w_2^2 \leq r^2$ forms a perfect **circle** centered at the origin. The radius r is inversely proportional to λ .
- The OLS loss function forms **elliptical** contour lines expanding outward from its global minimum (which usually sits far outside the circle).

- **The intersection:** The solution lies on the **boundary (perimeter)** of the constraint circle where the OLS contour ellipse barely touches it (tangent point).



Because the solution is forced to lie on the boundary or "**ridge**" of this constraint circle, the technique earned the name **Ridge Regression**.

4.6 A Practical Heuristic

Ridge Regression becomes genuinely useful when the number of input features is $N \geq 2$. With only one feature (simple linear regression), applying Ridge is rarely beneficial. The true power of L2 regularization is managing the complex interplay, multicollinearity, and overfitting potential in high-dimensional, multi-feature spaces.

★ KEY INSIGHT

Ridge shrinks all coefficients toward zero but never makes them exactly zero. Larger coefficients are penalized more aggressively. The constraint region is a circle, and the name "Ridge" comes from the solution lying on the boundary (ridge) of that circle.

◆ TAKEAWAY

Six key properties: (1) Coefficients never reach exactly zero. (2) Larger coefficients shrink faster. (3) Bias increases and variance decreases with λ . (4) The loss function parabola shifts toward the origin. (5) The circle constraint explains the name "Ridge." (6) Ridge is most useful with $N \geq 2$ features.

PRACTICE P1 · RIDGE REGRESSION

(a) A student trains a Linear Regression model on a housing dataset with 30 features. The training RMSE is 1,200 but the test RMSE is 15,800. The top three coefficients are $w_5 = 8400$, $w_{12} = -6200$, and $w_{23} = 5100$.

(i) Is this model likely overfitting or underfitting? What evidence supports your answer?

(ii) If Ridge Regression is applied with a well-tuned λ , what do you expect to happen to the training RMSE and the test RMSE?

(iii) Will the coefficients w_5, w_{12}, w_{23} remain at their current values, increase, or decrease in magnitude after Ridge is applied?

(b) Given:

$$X^T X = \begin{bmatrix} 4 & 2 \\ 2 & 4 \end{bmatrix}, \quad X^T y = \begin{bmatrix} 10 \\ 6 \end{bmatrix}, \quad \lambda = 2$$

- (i) Write and compute $(X^T X + \lambda I)$: _____
 (ii) Compute $\beta_{\text{ridge}} = (X^T X + \lambda I)^{-1} X^T y$: _____
 (iii) Now compute $\beta_{\text{OLS}} = (X^T X)^{-1} X^T y$: _____
 (iv) Compare: which set of coefficients has a smaller total squared magnitude ($\sum \beta_j^2$)? Why does this make sense?

PRACTICE P1-B · RIDGE REGRESSION (Homework)

- (a) A model has OLS coefficients $w_1 = 200$ and $w_2 = 3$. You apply Ridge with a moderate λ .
 (i) Which coefficient will shrink more in absolute terms? Explain using the penalty mechanism.

 (ii) After regularization, could w_2 end up larger than w_1 ? Explain.

 (d) **Concept check: The denominator argument.** The Ridge formula for a 2D slope is:

$$m_{\text{Ridge}} = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2 + \lambda}$$

A student says: "If I set $\lambda = 10^{12}$, the coefficient will be exactly zero." Is this correct? Why or why not?

PART III: L1 REGULARIZATION — LASSO REGRESSION

Lasso Regression: L1 Regularization

▷ HOOK

Ridge Regression pulls coefficients closer to zero but never exactly makes them zero. What if we want a model that not only controls coefficients but also identifies which features are truly important and automatically discards the rest?

A quick analogy

Ridge is like turning down the volume on every speaker in a room — everyone still speaks, but more softly. No one is removed. **Lasso** is like a talent show judge — only the best performers stay on stage. Everyone else is sent home (coefficient = 0).

Keep this analogy in mind as we formalize Lasso mathematically.

The mathematical formulation

Lasso (Least Absolute Shrinkage and Selection Operator) uses a different penalty. Instead of the squared magnitude, Lasso penalizes the absolute magnitude of the coefficients.

$$L_{\text{Lasso}} = \underbrace{\sum_{i=1}^n (y_i - \hat{y}_i)^2}_{\text{OLS error}} + \lambda \underbrace{\sum_{j=1}^p |w_j|}_{\text{L1 penalty}}$$

Here:

- w_j : the coefficients (slopes) of the input features
- λ (or α): the hyperparameter controlling the strength of regularization, $\lambda \geq 0$

The impact of increasing λ

λ value	Coefficient behavior	Model state
$\lambda = 0$	Standard OLS values; potentially large and overfitting	Overfitting
Intermediate λ	Less important features hit exactly zero; remaining shrink	"Sweet spot" for feature selection
Very high λ	All coefficients forced to zero; flat line ($\hat{y} = c$)	Underfitting

◆ TAKEAWAY

Lasso uses L1 penalty ($\lambda \sum |w_j|$). The critical behavioral difference from Ridge emerges in the next section.

Key Differences: Lasso vs. Ridge Regression

▷ HOOK

The change from w_j^2 (L2 norm) to $|w_j|$ (L1 norm) might seem minor mathematically, but it leads to profoundly different behavior in the resulting models.

Feature Selection and Sparsity

The most critical difference is that **Lasso Regression can drive the coefficients of less important features exactly to zero.**

- **Ridge Regression:** Shrinks coefficients, keeping all features in the model (e.g., $w_1 = 0.5, w_2 = 0.001$).
- **Lasso Regression:** Acts as an automatic **Feature Selection** mechanism. If a feature does not significantly contribute to predicting the target variable y , Lasso forces its coefficient to 0.

When multiple coefficients become exactly zero, the resulting weight vector is called **sparse**. Lasso Regression is famous for creating **sparsity** in models.

When to use Lasso vs. Ridge?

Scenario	Preferred method	Reason
High-dimensional data with many irrelevant features	Lasso	Drops irrelevant columns, simplifies model, reduces dimensionality, easier to interpret
Domain knowledge says all features contribute	Ridge	Keeps all features with reduced influence, preserves all information

★ KEY INSIGHT

Ridge keeps all features and only reduces their magnitudes. Lasso eliminates features entirely. This makes Lasso a dimensionality reduction tool and Ridge a coefficient stabilization tool.

◆ TAKEAWAY

Ridge and Lasso represent different beliefs about the coefficient structure. Ridge assumes all features contribute (dense model). Lasso assumes many features are irrelevant (sparse model). Neither is universally better — the choice depends on the problem.

Optimization note: Ridge has a closed-form solution (one-step matrix computation), while Lasso requires iterative numerical optimization (e.g., coordinate descent) because the L1 penalty creates a non-differentiable kink at zero.

Geometric and Visual Intuition of Lasso

▷ HOOK

We now understand what Lasso does differently. But why does it happen? The answer lies in the geometry of the L1 penalty — both in the shape of the loss function and the shape of the constraint region.

Geometric interpretation: Diamond vs. Circle

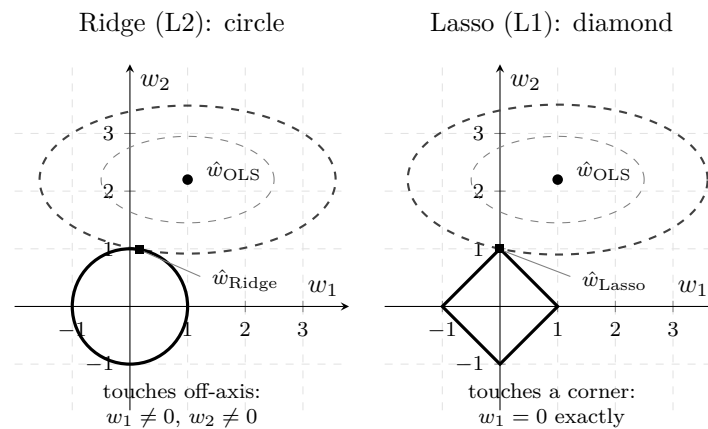
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Lasso constrains the coefficients to lie within:

$$\sum_{j=1}^p |w_j| \leq t$$

In two dimensions, this is a **diamond** (rotated square) centered at the origin.

Method	Constraint shape	Why the shape matters
Ridge	Circle (smooth boundary)	RSS contours touch the smooth surface at a non-axis point — no coefficient is forced to zero
Lasso	Diamond (sharp corners on axes)	RSS contours are likely to first touch at a corner — where one or more coefficients are exactly zero



Bias–Variance tradeoff in Lasso

Just like Ridge, Lasso modulates the Bias–Variance Tradeoff:

- At **low** λ : model is complex. Low Bias, High Variance (overfitting).
- As λ **increases**: penalty introduced. Variance drops significantly, Bias slowly climbs.
- At **very high** λ : all weights become zero. Extremely high Bias (underfitting).

The goal is to use cross-validation to select a λ in the middle zone.

◆ TAKEAWAY

Lasso's L1 penalty creates a "V"-shaped loss at $w = 0$ and a diamond-shaped constraint region. The diamond's sharp corners on the axes cause RSS contours to touch at vertices, forcing coefficients to exactly zero.

Why does Lasso create sparsity? (Mathematical proof)

▷ HOOK

A classic ML interview question: "Why does Lasso (L1) force coefficients exactly to zero, while Ridge (L2) only pushes them close to zero?" The answer lies in where λ appears in the coefficient formula.

The derivative of the Lasso loss function

Consider a single feature x , slope m . The Lasso loss is:

$$L_{\text{Lasso}} = \sum_{i=1}^n (y_i - mx_i - c)^2 + \lambda|m|$$

Because $|m|$ is not differentiable at $m = 0$, we analyze using a piecewise approach.

Let $S_{xy} = \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$ and $S_{xx} = \sum_{i=1}^n (x_i - \bar{x})^2$.

Case 1: When $m > 0$

Here $|m| = m$. Taking $\frac{\partial L_{\text{Lasso}}}{\partial m} = 0$:

$$-2S_{xy} + 2mS_{xx} + \lambda = 0 \implies m = \frac{S_{xy} - \frac{\lambda}{2}}{S_{xx}} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) - \frac{\lambda}{2}}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

Notice that λ appears in the numerator as a **direct subtraction**.

- Suppose $S_{xy} = 100$ and $S_{xx} = 40$.
- If $\lambda = 40$: $m = \frac{100-20}{40} = 2 > 0$ ✓
- If $\lambda = 200$: $m = \frac{100-100}{40} = 0$

- If $\lambda = 240$: formula gives $\frac{100-120}{40} = -0.5 < 0$, violating $m > 0$. The optimization clips at $m = 0$.

Case 2: When $m < 0$

Here $|m| = -m$. Taking $\frac{\partial L_{\text{Lasso}}}{\partial m} = 0$:

$$-2S_{xy} + 2mS_{xx} - \lambda = 0 \implies m = \frac{S_{xy} + \frac{\lambda}{2}}{S_{xx}} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) + \frac{\lambda}{2}}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

λ is **added** to the numerator. As λ increases, a negative m is driven toward zero until it clips at $m = 0$.

Comparing with Ridge Regression's formula

In Ridge Regression, the exact derivative yields:

$$m_{\text{Ridge}} = \frac{S_{xy}}{S_{xx} + \lambda} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2 + \lambda}$$

λ sits in the **denominator**. No matter how large λ becomes (e.g., 10^9), you are dividing the numerator by a massive number. The result becomes infinitesimally small (e.g., 10^{-9}) but mathematically **never** reaches exactly 0.

★ KEY INSIGHT

- **Lasso (L1):** λ in the **numerator** (subtraction) $\rightarrow$ coefficient is forced to exactly zero once λ surpasses the feature's correlation with the target.
- **Ridge (L2):** λ in the **denominator** (division) $\rightarrow$ coefficient shrinks asymptotically but never reaches exactly zero.

◆ TAKEAWAY

L1 creates sparsity because λ subtracts directly in the numerator. L2 does not create sparsity because λ only scales the denominator. This is the mathematical reason why Lasso performs feature selection and Ridge does not.

PRACTICE P2 · LASSO, SPARSITY, AND COMPARISON

(a) Application: Feature importance. A hospital uses 50 features to predict patient readmission. After applying Lasso with cross-validated λ , 42 coefficients become zero and only 8 survive.

- What term describes a model with many zero coefficients? _____
- A colleague suggests using Ridge instead "because it is more stable." If Ridge were applied, approximately how many features would remain in the model? _____. Explain why.

- From a clinical interpretability standpoint, which model (the 8-feature Lasso or the 50-feature Ridge) would be easier for doctors to explain to patients? Why?

(b) Concept check: Mathematical proof of sparsity. Consider the exact Lasso formula for $m > 0$: $m = \frac{S_{xy} - \lambda/2}{S_{xx}}$.

Suppose $S_{xy} = \sum (x_i - \bar{x})(y_i - \bar{y}) = 100$ and $S_{xx} = \sum (x_i - \bar{x})^2 = 40$.

- For $\lambda = 40$, compute m : _____
- For $\lambda = 200$, compute m : _____
- For $\lambda = 240$, what does the formula evaluate to? _____. What value does the Lasso algorithm actually output? _____

Now consider the exact Ridge formula: $m = \frac{S_{xy}}{S_{xx} + \lambda} = \frac{100}{40 + \lambda}$.

- For $\lambda = 10^6$, compute m : _____. Is it exactly zero? _____

(c) Application: Choosing the right method. For each scenario, choose Ridge or Lasso and give one reason.

Scenario	Choice	Reason
Gene expression data with 10,000 genes, most expected to be irrelevant	_____	_____
Predicting sales using 5 well-understood economic indicators	_____	_____
20 highly correlated sensor readings from a factory	_____	_____
Customer churn model with 200 features; business wants a simple scorecard	_____	_____

(d) Geometric reasoning. A student draws two constraint regions on the same axes — a circle (Ridge) and a diamond (Lasso). The OLS solution is at $(3, 0.5)$.

(i) Which constraint region is more likely to produce a solution where $w_2 = 0$? _____

(ii) Explain geometrically why this happens:

(e) Penalty calculation. Given the coefficient vector $\mathbf{w} = [5, -2, 8]$, compute:

(i) The L1 penalty ($\sum |w_j|$): _____

(ii) The L2 penalty ($\sum w_j^2$): _____

(iii) Which penalty would a Lasso model use? _____ Which would Ridge use? _____

PART IV: SIDE-BY-SIDE COMPARISON AND PRACTICAL CONSIDERATIONS

Complete comparison and practical usage

Side-by-side comparison

Aspect	Ridge (L2)	Lasso (L1)
Penalty term	$\lambda \sum w_j^2$	$\lambda \sum w_j $
Shrinkage	Toward zero, never exactly zero	Some coefficients become exactly zero
Feature selection	No — keeps all features	Yes — automatic feature selection
λ position in formula	Denominator (scaling)	Numerator (subtraction)
Constraint geometry	Circle (smooth)	Diamond (sharp corners)
Loss function shape	Smooth parabola	"V"-shaped kink at zero
Closed-form solution	Yes: $(X^\top X + \lambda I)^{-1} X^\top y$	No — requires numerical optimization
Best when	All features contribute; multi-collinearity	Many features irrelevant; want interpretable model

Cross-validation for λ selection

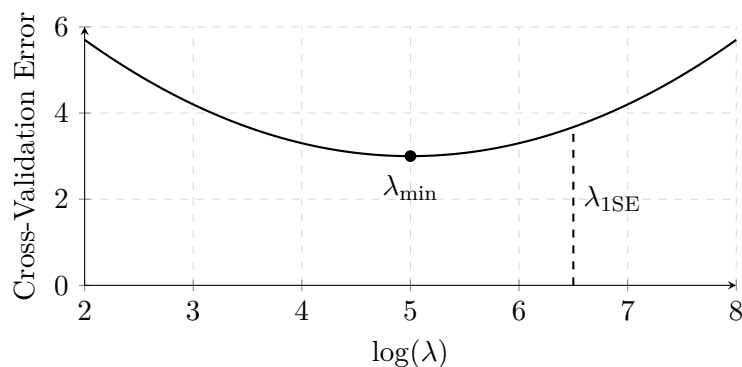
To find the optimal regularization strength λ :

1. Evaluate a candidate grid of λ values using k -fold cross-validation.
2. Select $\lambda_{\min}$ (lowest validation error) or λ_{1SE} (simplest model within 1 std error of minimum).

Classroom Exercise: A team tests four λ values. Which should they select?

λ	0	0.1	1	10
Avg. Validation MSE	150	85	72	310

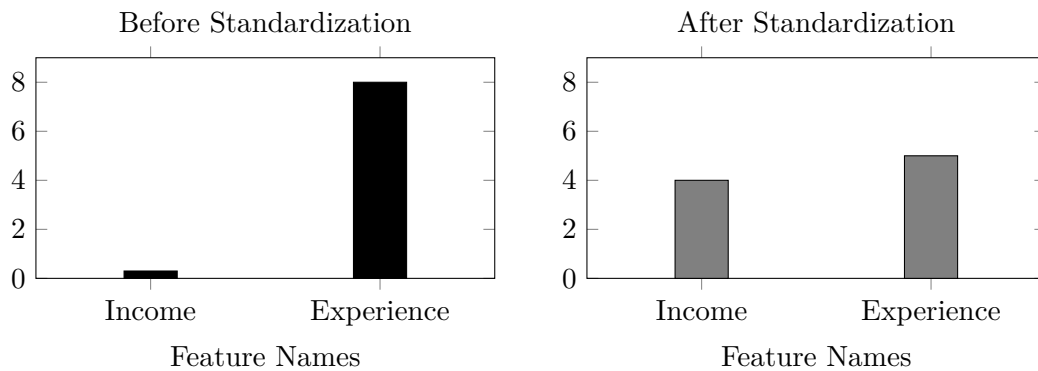
Best λ : _____ Reason: _____



Standardization before regularization

Because regularization penalizes raw coefficient magnitudes (w_j), features on smaller scales have naturally larger coefficients and get penalized far more heavily.

Rule: Standardize all features ($x_j^{\text{std}} = \frac{x_j - \mu_j}{\sigma_j}$) before fitting. Do **not** regularize the intercept term (β_0).



★ KEY INSIGHT

Standardization ensures the penalty treats all features equally regardless of measurement scales. Without it, a feature in thousands (income in rupees) would be penalized far more than one in single digits (years of experience).

◆ TAKEAWAY

Ridge keeps all features with reduced coefficients; Lasso eliminates unimportant ones. λ is selected via cross-validation. Features must be standardized so the penalty is fair. The intercept is not regularized.

Common Mistakes to Avoid

Mistake	Why it's wrong
Regularizing the intercept (β_0)	The intercept shifts the prediction baseline — penalizing it biases the model toward $\hat{y} = 0$ instead of $\hat{y} = \bar{y}$
Standardizing before train-test split	This leaks test-set statistics into training, producing overly optimistic results (data leakage)
Assuming larger λ is always better	Too large λ causes underfitting — the optimal λ is found via cross-validation
Believing Ridge performs feature selection	Ridge shrinks but never zeros out coefficients — only Lasso performs feature selection
Using raw (unstandardized) features	The penalty unfairly targets features with naturally large coefficients due to measurement scale

Interview Traps — True or False?

Statement	T / F	Reason
"Ridge Regression performs feature selection."	_____	_____
"Ridge coefficients can become exactly zero if λ is large enough."	_____	_____
"Lasso always outperforms Ridge."	_____	_____
"The intercept (β_0) should be regularized along with other coefficients."	_____	_____

PRACTICE P3 · APPLICATION AND SYNTHESIS

(a) Debugging a pipeline. A data science team applies Lasso to predict crop yield using 4 features:

- Rainfall (mm, range 200–2000)
- Temperature ($^{\circ}\text{C}$, range 10–45)
- Soil pH (range 4.5–8.0)
- Fertilizer cost (Rs., range 5000–50000)

After training, Lasso sets the coefficients for Rainfall and Soil pH to zero, keeping only Temperature and Fertilizer cost. A domain expert is surprised: "Rainfall is definitely important for crop yield!"

- (i) What preprocessing step did the team most likely skip? _____
- (ii) Why would skipping this step cause Rainfall's coefficient to be unfairly treated by the penalty?

- (iii) After fixing the issue, would you expect the same features to be eliminated? Explain.

(b) Interpreting regularization paths. Below are two regularization path descriptions.

Path A: As λ increases from 0 to 100, all 10 coefficient curves smoothly approach zero but none touches the x-axis.

Path B: As λ increases from 0 to 100, coefficient curves shrink and hit zero one by one. At $\lambda = 100$, only 3 features remain non-zero.

- (i) Which path corresponds to Ridge? _____ Which to Lasso? _____
- (ii) At $\lambda = 100$, how many features does each model effectively use?

Path A: _____ Path B: _____

(c) Sequence arrangement. A colleague sends you an unordered list of steps for building a Ridge model. Put them in order.

☐ Compute $(X^T X + \lambda I)^{-1} X^T y$ ☐ Split data into training and test sets ☐ Fit mean μ_j and std σ_j on training set only ☐ Select λ via cross-validation ☐ Standardize features using training-set statistics ☐ Evaluate on the held-out test set

Correct order: _____ $\rightarrow$ _____ $\rightarrow$ _____ $\rightarrow$ _____ $\rightarrow$ _____

Why must you fit μ_j and σ_j only on the training set?

(d) Interview question: Why is the intercept not regularized?

In the Ridge loss function, the penalty is $\lambda \sum_{j=1}^p w_j^2$ — the summation starts at $j = 1$, deliberately excluding the intercept w_0 (or β_0). Why?

PRACTICE P4 · COMPREHENSIVE CHALLENGE (Homework)

(a) Case study. An e-commerce company wants to predict customer lifetime value (CLV) using 80 features. The data scientists know from experience that only about 10–15 features actually drive CLV. The remaining features are mostly marketing campaign IDs, timestamp encodings, and other noisy variables.

(i) Should they use Ridge or Lasso? Justify your choice.

(ii) After applying the chosen method, the team finds training $R^2 = 0.97$ and test $R^2 = 0.94$. The λ was selected via cross-validation. Is this a good model? Explain.

(iii) A manager asks: "Why not just set $\lambda = 0$ to get the best training accuracy?" Write a brief response using the Bias–Variance Tradeoff.

(b) Error identification. A student writes the following Ridge Regression cost function:

$$J(\beta) = \sum_{i=1}^n (y_i - \hat{y}_i)^2 + \lambda \sum_{j=0}^p |\beta_j|$$

Identify and explain all errors (there are at least two).

(c) Conceptual scenario. Two colleagues are debating:

Colleague A: "I used Ridge with $\lambda = 1000$ and all my coefficients are zero. Ridge can produce zero coefficients if λ is large enough."

Colleague B: "That's impossible. Ridge coefficients never become exactly zero no matter how large λ is."

Who is correct? Explain using the mathematical formula.

(d) Synthesis questions. Answer any three.

(i) Geometrically, why does Lasso produce zero coefficients but Ridge does not? (Use the circle vs. diamond argument.)

(ii) Mathematically, why does L1 create sparsity but L2 does not? (Use the numerator vs. denominator argument.)

(iii) A model uses Lasso and sets λ very large. Describe the resulting model in terms of bias, variance, and number of features used.

(iv) How does the regularization parameter λ relate to the bias–variance tradeoff? Describe the behavior at $\lambda = 0$, optimal λ , and $\lambda \rightarrow \infty$.

(v) Why must features be standardized before applying regularization? Give a concrete example.

(vi) A company has 15 features: 5 are known to be highly predictive and the rest are noisy. Recommend a regularization method and explain your reasoning.

(e) Evaluate: Defend your recommendation. A pharmaceutical company is building a model to predict drug efficacy using 200 molecular features. The business team requires an interpretable model — regulators need to know *which* molecular properties drive efficacy, and the model must be explainable in a submission report.

Should the team use Ridge or Lasso? Defend your choice by addressing: (1) interpretability, (2) number of features, and (3) regulatory requirements.

BY THE END OF THIS SHEET, YOU CAN

- ☐ Explain why overfitting produces abnormally large coefficients and why regularization is needed.
- ☐ Write the Ridge loss function ($L_{\text{OLS}} + \lambda m^2$), explain the bias–variance compromise, and derive the 2D closed-form solution.
- ☐ Describe six key properties of Ridge: never-zero coefficients, asymmetric shrinkage, bias–variance impact, geometric shifting, circle constraint ("Ridge" name), and practical heuristic.
- ☐ Write the Lasso loss function and explain the key differences from Ridge: feature selection, sparsity, and when to use each.
- ☐ Explain the geometric intuition: diamond constraint with sharp corners vs. Ridge's circular constraint.
- ☐ Prove mathematically why L1 creates sparsity (numerator) while L2 does not (denominator).
- ☐ Select λ via cross-validation, explain standardization, and decide which method suits a given problem.

Worksheet 12 · AML · Lecture 12: Regularization (L1 and L2).